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Advanced PLC Programming

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Advanced PLC Project Proposal

I will be the only person in my team. My project is going to be to automate the MPO and SLD, and be able to control those two systems with an HMI. I will include safety devices and event logging. The safety devices will all be local to the HMI and PLC, but the event logging will be done through a Python application running on my laptop. To adjust the project for finals, I will be implementing scheduled scripts to grab data from both of the modules periodically using Systemd, as well as developing a website which the user will be able to utilize to look at the databases and SCADA information. This will be implemented through a SQLite database utilizing Django on the backend with HTML, CSS, and Javascript on the frontend. My timeline is as follows:

Feb 1 – 16: Program the MPO and SLD on the PLC, keeping safety mechanisms such as E-Stop and fusing in mind. Also program the HMI during this time.

Feb 17 – 28: Get the PyModbus and database application working.

March 1 – 13: Connecting the HMI + PLC to the PyModbus application, debugging.

March 23 – April 3: Understand Systemd scheduling

April 4 – 17: Update PyModbus application and database to be able to grab data from the factories and store it. Develop the frontend website

April 18 – May 1: Connect the new website and database to the PLC and HMI.

May 1 – 15: Debug any issues which may arise.

The midterm project requirements which I will have are:

- Safety Devices with your system (These will be done with the HMI and in software)
- HMI event logging (To a database application which I will develop)

The final project requirements which I will have are:

- A localized front-end website that displays features and functionality of your system for SCADA
(Django + SQLite + HTML + CSS + Javascript)
- Systemd service for running command scripts that grab your data at regular intervals from your system.